# Performance of Natural Language Processing models in extracting important clinical information from a free text radiology report

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**File name** : 24108_102.txt
**Original file size** : 346.31 KB
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# *23CSE301 – Machine Learning*

# *ASSIGNMENT*

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**Paper Title** – *Performance of Natural Language Processing models in extracting important clinical information from a free text radiology report*



## 1. Problem Understanding

Background of the Problem

Radiology reports contain important clinical information about a patient's condition. These reports are commonly written as free-text, where radiologists describe their observations and findings in their own words. this format is useful for doctors because it gives them the freedom to explain the findings in detail and in their own way But this same flexibility makes it difficult for computers to understand and process the report properly Many useful details like abnormalities anatomical locations and diagnostic findings are present in the report but they are not available in a structured format Because of this it becomes difficult to use the large amount of radiology data in an efficient way As the number of radiology reports increases this problem becomes more important Different radiologists can also use different words or expressions for describing the same finding and some reports may contain repeated information while some important details may be missing This makes it more difficult to automatically find and extract the required information If the information is extracted manually each report has to be read and checked which takes more time and can also cause some errors So the main idea of this work is to use NLP to identify the important clinical information from free text radiology reports and convert it into a more structured form which can be used for further analysis

Role of NLP in Solving the Problem

Natural Language Processing (NLP) can be used to solve this problem by allowing computers to process and understand medical text. NLP models can identify useful information from free-text radiology reports, such as clinical findings and

anatomical information, and extract them in a structured form. Different NLP methods are used for this problem in the base paper such as rule based methods machine learning methods deep learning methods and Large Language Models In the earlier methods the models mostly depended on manually written rules selected features and annotated data Later language models were used to work with medical text with less dependence on manually created rules The base paper does not develop only one NLP model Instead it looks at different research studies and compares their results through a systematic review and meta analysis I understood from this that the main aim of the paper is to see how well different NLP methods are able to take useful information from free text radiology reports

Free-text Radiology Report → Difficult for computers to process → NLP extracts important information → Structured Clinical Information



Finding Status Location

Pneumonia Present Right lower lobe

Pleural effusion Absent Not applicable

## 2. Dataset Description – What Information I Got from the Dataset

For my project I selected the OpenI Chest X ray Dataset from Indiana University It is a public dataset and the patient information has been de identified so direct details of the patients are not included The dataset contains 3955 radiology reports along with 7470 chest X ray images For my project I am mainly using the text part of the dataset because my work is focused on extracting useful information from radiology reports and not on analysing the X ray images. In each radiology report is divided into different sections. The main sections available are Comparison, Indication, Findings, and Impression. Among these, the Findings and Impression sections are the most useful for my work. The Findings section contains what the radiologist observed in the X-ray, while the Impression section contains the final conclusion or summary given by the radiologist.

For example, a report may contain a sentence such as:

> "Mild right lower lobe pneumonia. No pleural effusion."

From this text, the model should be able to understand that pneumonia is present, its location is the right lower lobe, and pleural effusion is absent. This is basically the type of information extraction that my project is trying to perform.

The dataset gives me different types of information that can be used for training and testing the NLP model.

• Radiology report text: This is the main input for my project.

• Findings: Describes the observations made by the radiologist.

• Impression: Gives the diagnostic conclusion.

• MeSH and RadLex terms: These provide medical terminology associated with the reports.

• X-ray images: These are available in the dataset, but I am not using them because my project focuses on extracting information from text.



The project uses the CheXpert label schema which contains 14 different chest related findings For each finding the result can be Positive Negative or Uncertain Since one radiology report can have more than one finding at the same time the problem is treated as a multi label classification task I selected this dataset because it gives me actual radiology report text and it is publicly available with patient details removed so I can work with the data without depending on private hospital records Another reason for selecting this dataset is that my project is mainly focused on chest radiology So instead of trying to work

with many different parts of the body I am keeping the problem limited to chest related findings This also connects with the base paper where extraction from a single anatomical site had an AUC of 0.99 while extraction from multiple anatomical sites had an AUC of 0.95 For me this makes the problem more focused and also gives a clear way to test the NLP model on one specific type of medical report, In simple terms the dataset flow for my project is

Radiology Report → Findings and Impression → Medical Text → Chest Finding Labels

My main goal is to take this text as input and make the NLP model understand which chest findings are present, absent, or uncertain in the report.

### 3. Approach Used in the Paper and Its Interpretation

The base paper uses a systematic review and meta analysis approach to study how NLP models are being used for extracting information from free text radiology reports. The paper does not train one particular model and test it on one dataset Instead the authors collected results from previous research studies and analysed them together Base Paper. The main idea was to understand the overall performance of NLP instead of depending on the result of only one study.

The authors searched different research databases to find studies related to NLP based information extraction from radiology reports The databases included PubMed MEDLINE Embase EBSCO Ovid Web of Science and the Cochrane Library The search was completed between November 21 and 23 2024 .After screening the available studies 28 studies were selected for the final analysis These studies included 51,187 free text radiology reports and 421,692 extracted clinical entities .This allowed the authors to study the problem using results from many different research works rather than depending on one hospital or onedataset

NLP Methods in the Paper

The studies included in the review used different types of NLP approaches

• Rule Based NLP



• Machine Learning Models

• Deep Learning Models

• Large Language Models

Traditional NLP methods mainly depended on manually created rules features and annotated data Later with the development of deep learning more advanced models started being used for extracting information from medical reports Large Language Models were also introduced as another way of working with medical text I understood that the methods slowly changed from using fixed rules and manually prepared features to models which can understand more of the context in the text. For each study the authors collected details such as the NLP model used number of radiology reports number of extracted entities dataset source imaging modality language anatomical site and whether external validation was done They also collected diagnostic values such as true positives false positives false negatives and true negatives If these values were not directly given in a study the authors derived them from other reported performance measures when it was possible The main idea was to bring the results from different studies into one common format so they could be compared and analysed together. After collecting all the required information the authors performed a meta analysis using a bivariate random effects model This method was used for analysing diagnostic accuracy by considering sensitivity and specificity together This helped the authors combine the results from different studies and get an overall view of how well the NLP methods were performing

The overall results reported in the paper were as follows

Sensitivity 91 percent

• Specificity 96 percent

• Diagnostic Odds Ratio 220

• AUC 0 point 98 Base Paper

These results show that NLP systems were able to extract useful clinical information from free text radiology reports with high overall performance

Grouping represent :-

Those authors also checked whether the performance changed under different conditions They looked at factors such as NLP model type dataset source external validation language imaging modality and anatomical site One important result was related to the anatomical site Extraction from a single anatomical site had an AUC of 0.99 while extraction from multiple anatomical sites had an AUC of 0.95 The paper also found that anatomical site was the only significant factor contributing to the variation between the studies in the multivariable meta regression



This part is important for my project because I am focusing only on chest radiology instead of trying to work with many different anatomical regions at the same time

From my understanding the main purpose of the paper was not to find one particular NLP model and call it the best The authors mainly wanted to understand how well NLP based information extraction was working across the available research The results showed that the overall performance was strong but there were still differences between the studies The paper also points towards the need for better standardisation of radiology reports and better model generalisation

In my understanding the base paper gives the research direction and my project is taking that idea and testing it on an actual public chest radiology dataset

Approach in My Project

The base paper studies the problem at a broader research level My project takes a more focused approach by using the OpenI Chest X ray Dataset and concentrating on chest radiology reports. The planned implementation uses the Findings and Impression sections as the main text input ClinicalBERT is planned for multi label classification and TF IDF with Logistic Regression and BioBERT are planned as comparison methods

The planned workflow is

Radiology Report → Text Preprocessing → ClinicalBERT → Chest Finding Prediction → Positive Negative or Uncertain

So the base paper gives the overall research idea and my project takes that idea into a practical implementation using a public chest radiology dataset

## 4. Conclusion

The main problem discussed in this work is the difficulty of extracting useful clinical information from free text radiology reports Since these reports are written in an unstructured way computers cannot directly use all the information present in them NLP provides a way to process this text and convert important information into a more structured form The base paper studied different NLP approaches including rule based methods machine learning deep learning and large language models Base PaperThe systematic review included 28 studies with 51 187 radiology reports and 421 692 extracted clinical entities The overall results showed a sensitivity of 91 percent specificity of 96 percent and an AUC of 0 point 98 Base Paper

From my understanding the paper shows that NLP can be a useful approach for making information present in radiology reports easier to extract and analyse



The paper also showed that performance can change depending on the type of task and anatomical site Single anatomical site

extraction achieved an AUC of 0 point 99 compared with 0 point 95 for multiple anatomical sites Base Paper. This is one of the main reasons I am focusing my project on chest radiology reports instead of trying to cover many anatomical regions

For my project the next step is to use the OpenI Chest X ray Dataset and work mainly with the Findings and Impression sections The planned approach is to use ClinicalBERT for multi label classification and compare its results with other baseline methods Overall the base paper gives me a clear starting point and my project aims to take the same problem into a practical implementation using a public dataset.